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Java SE 27 & JDK 27

Interface Iterable<T>

Type Parameters:

T - the type of elements returned by the iterator

All Known Subinterfaces:

BeanContext, BeanContextServices, BlockingDeque<E>, BlockingQueue<E>, ClassModel, ClassReader, CodeAttribute, CodeModel, Collection<E>, CompoundElement<E>, ConstantPool, ConstantPoolBuilder, Deque<E>, DirectoryStream<T>, EventSet, FieldModel, List<E>, MethodModel, NavigableSet<E>, NodeSetData<T>, Path, Queue<E>, SecureDirectoryStream<T>, SequencedCollection<E>, SequencedSet<E>, Set<E>, SortedSet<E>, TransferQueue<E>, XPathNodes

All Known Implementing Classes:

AbstractCollection, AbstractList, AbstractQueue, AbstractSequentialList, AbstractSet, ArrayBlockingQueue, ArrayDeque, ArrayList, AttributeList, BatchUpdateException, BeanContextServicesSupport, BeanContextSupport, ConcurrentHashMap.KeySetView, ConcurrentLinkedDeque, ConcurrentLinkedQueue, ConcurrentSkipListSet, CopyOnWriteArrayList, CopyOnWriteArraySet, DataTruncation, DelayQueue, DocTreePath, EnumSet, HashSet, JobStateReasons, LinkedBlockingDeque, LinkedBlockingQueue, LinkedHashSet, LinkedList, LinkedTransferQueue, PriorityBlockingQueue, PriorityQueue, RoleList, RoleUnresolvedList, RowSetWarning, SerialException, ServiceLoader, SQLClientInfoException, SQLDataException, SQLException, SQLFeatureNotSupportedException, SQLIntegrityConstraintViolationException, SQLInvalidAuthorizationSpecException, SQLNonTransientConnectionException, SQLNonTransientException, SQLRecoverableException, SQLSyntaxErrorException, SQLTimeoutException, SQLTransactionRollbackException, SQLTransientConnectionException, SQLTransientException, SQLWarning, Stack, SyncFactoryException, SynchronousQueue, SyncProviderException, TreePath, TreeSet, Vector

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`public interface` **`Iterable<T>`**

Implementing this interface allows an object to be the target of the enhanced `for` statement (sometimes called the "for-each loop" statement).

See *Java Language Specification*:

14.14.2 The enhanced `for` statement

Since:

1.5

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
`default void`	**`forEach`**`(Consumer<? super T> action)`	Performs the given action for each element of the `Iterable` until all elements have been processed or the action throws an exception.
`Iterator<T>`	**`iterator`**`()`	Returns an iterator over elements of type `T`.
`default Spliterator<T>`	**`spliterator`**`()`	Creates a `Spliterator` over the elements described by this `Iterable`.

Method Details

iterator

`Iterator<T> iterator()`

Returns an iterator over elements of type `T`.

Returns:

an Iterator.

forEach

`default void forEach(Consumer<? super T> action)`

Performs the given action for each element of the `Iterable` until all elements have been processed or the action throws an exception. Actions are performed in the order of iteration, if that order is specified. Exceptions thrown by the action are relayed to the caller.

The behavior of this method is unspecified if the action performs side-effects that modify the underlying source of elements, unless an overriding class has specified a concurrent modification policy.

Implementation Requirements:

The default implementation behaves as if:

```
    for (T t : this)
        action.accept(t);
```

Parameters:

`action` - The action to be performed for each element

Throws:

`NullPointerException` - if the specified action is null

Since:

1.8

spliterator

`default Spliterator<T> spliterator()`

Creates a `Spliterator` over the elements described by this `Iterable`.

Implementation Requirements:

The default implementation creates an *early-binding* spliterator from the iterable's `Iterator`. The spliterator inherits the *fail-fast* properties of the iterable's iterator.

Implementation Note:

The default implementation should usually be overridden. The spliterator returned by the default implementation has poor splitting capabilities, is unsized, and does not report any spliterator characteristics. Implementing classes can nearly always provide a better implementation.

Returns:

a `Spliterator` over the elements described by this `Iterable`.

Since:

1.8

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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